CHAPTER 2 - Introduction to LP

Ex. 2.1-1 The Reddy Mikks Company

Produces interior and exterior paints from raw materials M1 and M2

	Exterior	Interior	Daily Availability
M1	6	4	24
M2	1	2	6
Profit per ton ($1000s)	5	4	

- Daily demand for interior paint cannot exceed that of exterior by more than 1 ton. → $x_2 - x_1 \le 1$
- Maximum demand for interior is 2 tons. → $x_2 \le 2$
- What is the optimum product mix of exterior and interior which maximizes the total daily profit? ($x_1 = 1.5$, $x_2 = 1.5$)

All OR models, LP included, consist of three basic components:

- ✓ Decision variables? that we seek to determine (essential first step)
- ✓ Objective? (goal) that we need to optimize (max. or min.)
- ✓ Constraints? that the solution must satisfy.

Let, x_1 = amount of exterior paint produced daily

x_2 = amount of interior paint produced daily

The goal is … to maximize total daily profit of both paints.

Profit from ext. paint is $5x_1$ (thousand dollars)

Profit from int. paint is $4x_2$ "

maximize $z = 5x_1 + 4x_2$

s.t.

$$\begin{aligned} 6x_1 + 4x_2 &\le 24 \\ x_1 + 2x_2 &\le 6 \\ -x_1 + x_2 &\le 1 \\ x_2 &\le 2 \\ x_1, x_2 &\ge 0 \end{aligned}$$

constraints:

usage of a raw material by both paints ≤ maximum raw material availability

Usage of raw material M1 by both paints: $6x_1 + 4x_2 \le 24$

Usage of raw " M2 " : $x_1 + 2x_2 \le 6$

→ nonnegativity constraints.

mathematical model

- Any values of x_1 and x_2 satisfying all constraints is a feasible solution.

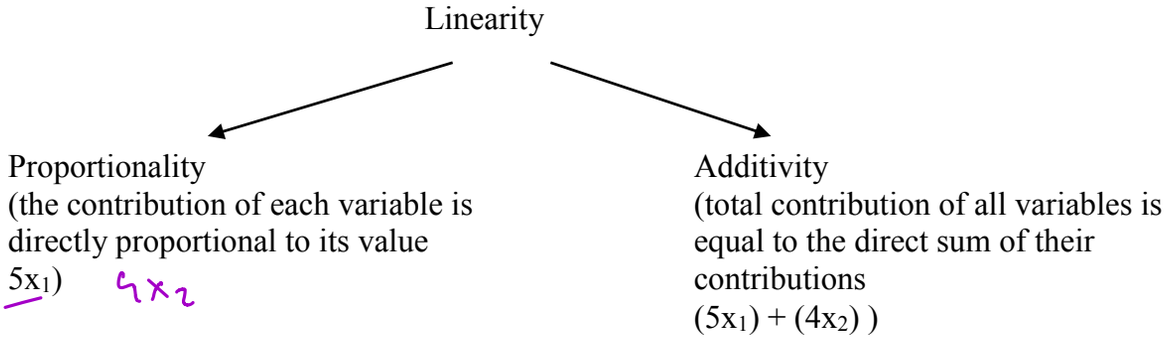
- The best among all feasible solutions is called the optimum solution.

The goal is to find the optimum.

Nontrivial LPs have infinite number of feasible solutions.

LP

In LP, the objective and constraints functions are linear.



Graphical Solution

$6x_1 + 4x_2 \leq 24 \Rightarrow x_1 = 0 \Rightarrow x_2 = 6, \quad x_2 = 0 \Rightarrow x_1 = 4$ → (0,6), (4,0)

$x_1 + 2x_2 = 6 \Rightarrow x_1 = 0 \Rightarrow x_2 = 3, \quad x_2 = 0 \Rightarrow x_1 = 6$ (0,3), (6,0)

$-x_1 + x_2 = 1 \Rightarrow x_1 = 0 \Rightarrow x_2 = 1, \quad x_2 = 0 \Rightarrow x_1 = -1$

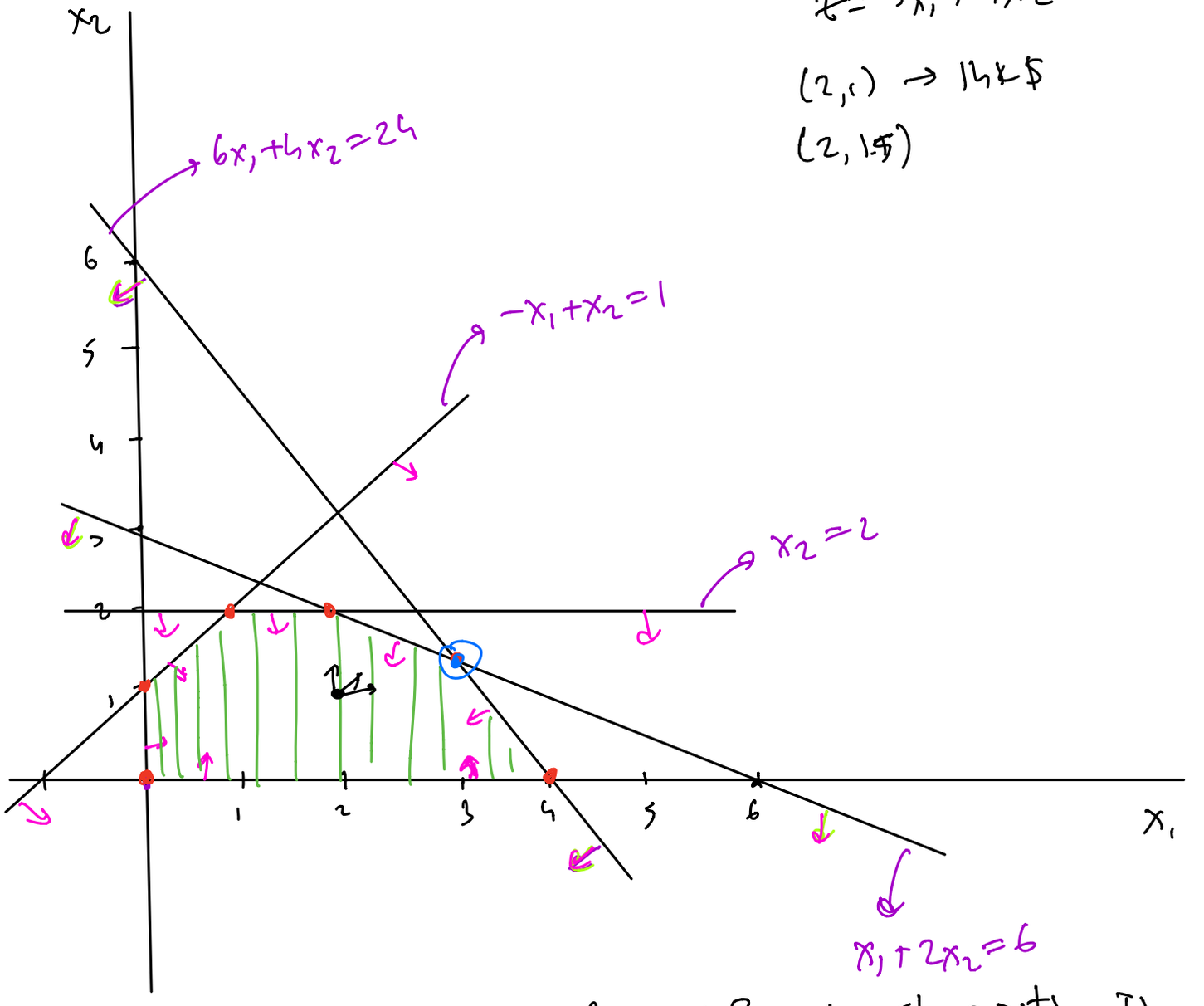
$x_2 = 2$

$x_1, x_2 \geq 0$

$z = 5x_1 + 4x_2$

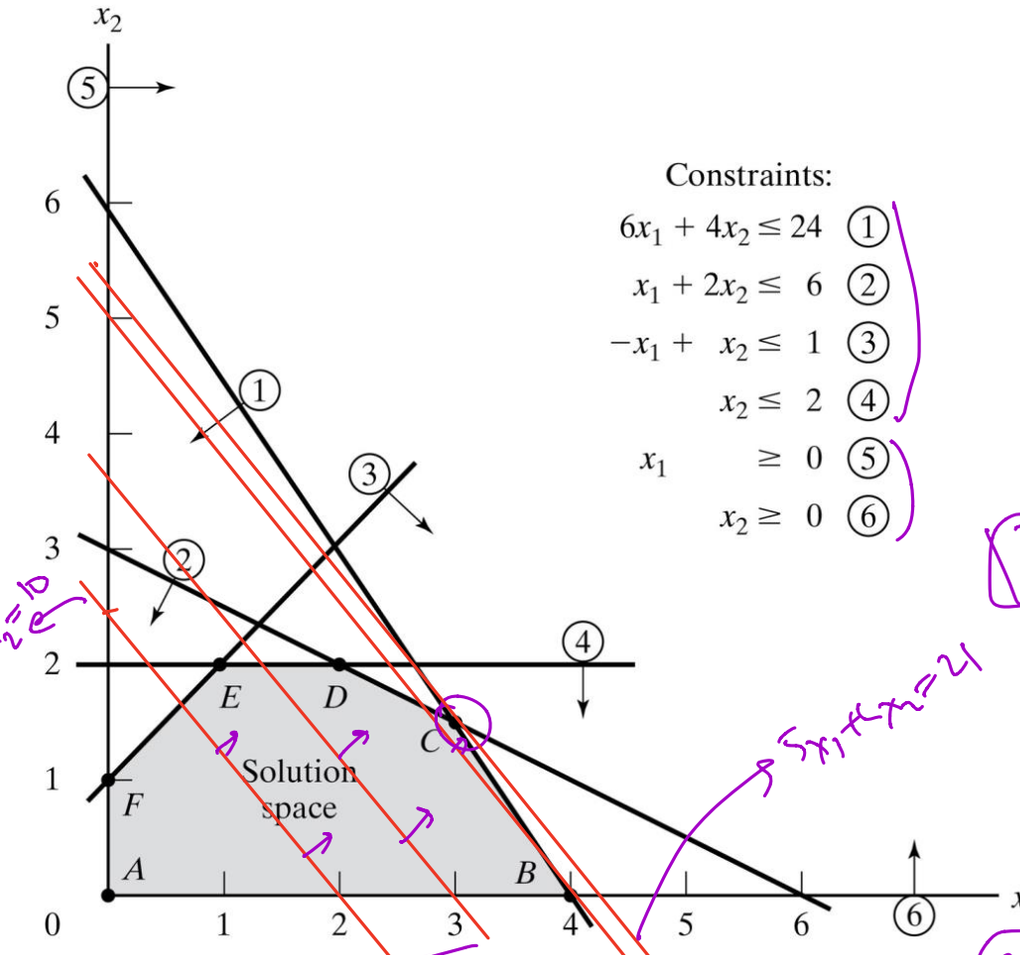
(2,1) → 14k$

(2, 1.5)



The optimum solution of an LP, when it exists, is always associated with a corner point of the solution space, thus limiting the search space for the optimum from an infinite number of feasible points to a finite number of corner points.

This is the basis of the simplex method.



Constraints:

$$6x_1 + 4x_2 \le 24 \quad (1)$$
$$x_1 + 2x_2 \le 6 \quad (2)$$
$$-x_1 + x_2 \le 1 \quad (3)$$
$$x_2 \le 2 \quad (4)$$
$$x_1 \ge 0 \quad (5)$$
$$x_2 \ge 0 \quad (6)$$

$5x_1 + 4x_2 = 10$

$x_1 = 0 \quad x_2 = 2.5$

$x_2 = 0 \quad x_1 = 2$

$5x_1 + 4x_2 = 15$

$x_1 = 0 \quad x_2 = 3.75$

$x_2 = 0 \quad x_1 = 3$

$$6x_1 + 4x_2 = 24$$
$$x_1 + 2x_2 = 6$$
$$4x_1 = 12 \Rightarrow x_1 = 3$$
$$x_2 = 1.5$$

$$z = 5x_1 + 4x_2 = 5 \times 3 + 4 \times 1.5 = 21$$